



INDIAN INSTITUTE OF TECHNOLOGY DELHI

CLL121: Thermodynamics

**Dynamic Optimization of Rankine Cycle
Performance under Variable Heat Input
Conditions**

Team Members

Gowtham Naik
Pranay Kumar
Aryan Raj

Date: 20/04/2026



Abstract

This project focuses on the performance analysis of the Rankine cycle using a simulation-based approach. The cycle is modeled using thermodynamic principles, and key operating parameters such as boiler pressure and turbine inlet temperature are varied to study their effects on system performance.

The thermal efficiency and net work output are evaluated under different conditions, and the results are represented graphically. The analysis shows that both efficiency and work output increase with an increase in pressure and temperature. The study provides a clear understanding of how operating conditions influence the performance of the Rankine cycle.

Contents

Abstract	2
1 Introduction	3
2 Objectives	3
3 Methodology	3
3.1 Rankine Cycle Model	3
3.2 Assumptions	4
4 Simulation Setup	4
5 Results and Discussion	4
6 Performance Analysis	7
7 Conclusion	7
8 Future Scope	7
9 References	7



1 Introduction

The Rankine cycle is a fundamental thermodynamic cycle widely used in thermal power plants to convert heat energy into mechanical work. It plays a key role in electricity generation systems where steam is used as the working fluid.

The performance of the Rankine cycle depends strongly on operating conditions such as boiler pressure and turbine inlet temperature. Variations in these parameters can significantly affect the efficiency and work output of the cycle.

In this project, a simulation-based approach is used to study the performance of the Rankine cycle under different operating conditions. By varying key parameters, the behavior of the system is analyzed and represented graphically to understand performance trends.

2 Objectives

- To model the Rankine cycle using thermodynamic principles
- To evaluate performance in terms of thermal efficiency and net work output
- To study the effect of boiler pressure and turbine inlet temperature
- To represent results graphically and analyze performance trends

3 Methodology

3.1 Rankine Cycle Model

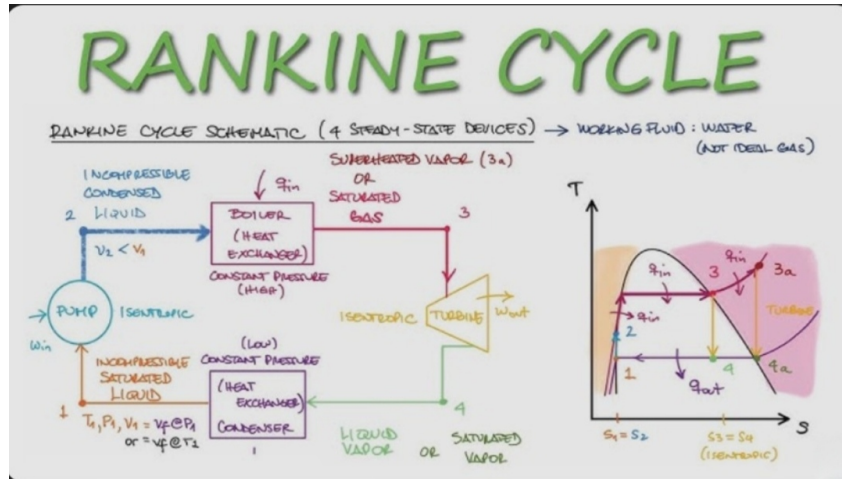


Figure 1: Schematic Diagram of Rankine Cycle (Source: Online)

The Rankine cycle consists of four main processes: pumping, heat addition in the boiler, expansion in the turbine, and heat rejection in the condenser. Water is first pumped to a higher pressure, then heated in the boiler to produce steam. The steam expands in the turbine to generate work, and is finally condensed back into liquid form in the condenser.



3.2 Assumptions

The following assumptions are made in the analysis of the Rankine cycle:

- The cycle operates under steady-state conditions
- The turbine expansion is isentropic (no entropy change)
- The pump work is considered negligible compared to turbine work
- Heat losses to the surroundings are neglected
- The working fluid follows standard steam table properties

4 Simulation Setup

The Rankine cycle is simulated using Python in Google Colab. Thermodynamic properties of steam are obtained using the pyXSteam library.

Two key parameters are varied in the simulation:

- Boiler pressure
- Turbine inlet temperature

For each variation, the thermal efficiency and net work output of the cycle are calculated. The results are then plotted to analyze performance trends.

5 Results and Discussion

The performance of the Rankine cycle is analyzed by varying boiler pressure and turbine inlet temperature. The results are presented in the form of graphs showing the variation of thermal efficiency and net work output.

From the graphs, it is observed that thermal efficiency increases with an increase in boiler pressure. This is because higher pressure increases the average temperature at which heat is added, resulting in improved energy conversion efficiency.

Similarly, the net work output of the cycle also increases with boiler pressure due to greater expansion of steam in the turbine.

When turbine inlet temperature is increased, both thermal efficiency and work output show an increasing trend. Higher temperature leads to higher enthalpy of steam at turbine inlet, which results in greater work output during expansion.

Overall, the results indicate that both boiler pressure and turbine inlet temperature have a significant effect on the performance of the Rankine cycle.

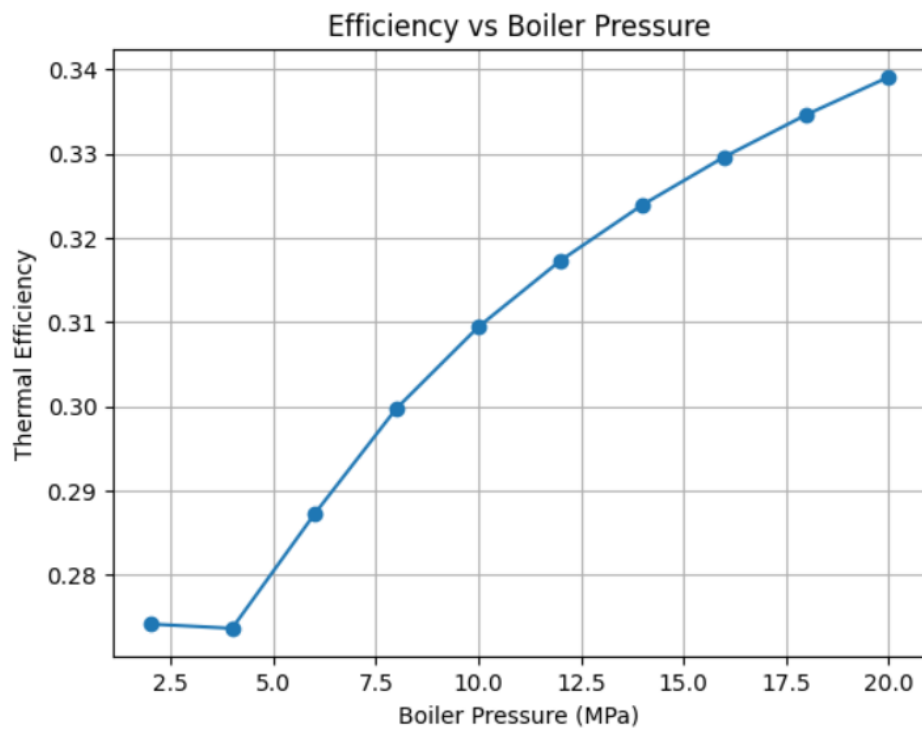


Figure 2: Efficiency vs Boiler Pressure

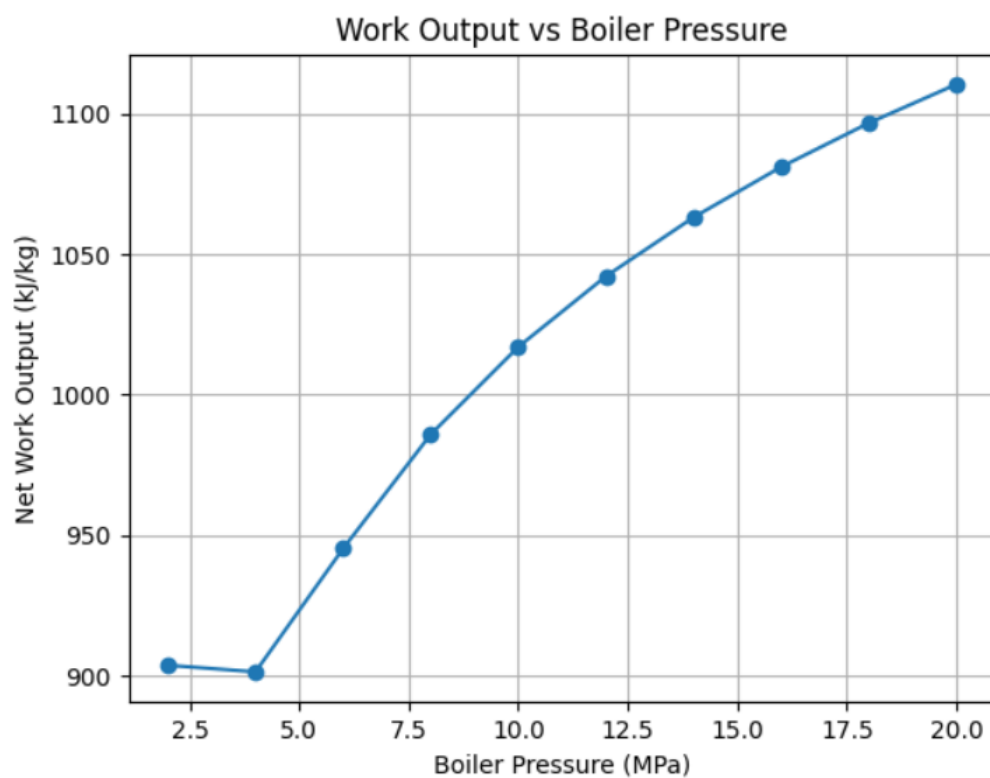


Figure 3: Work vs Boiler Pressure

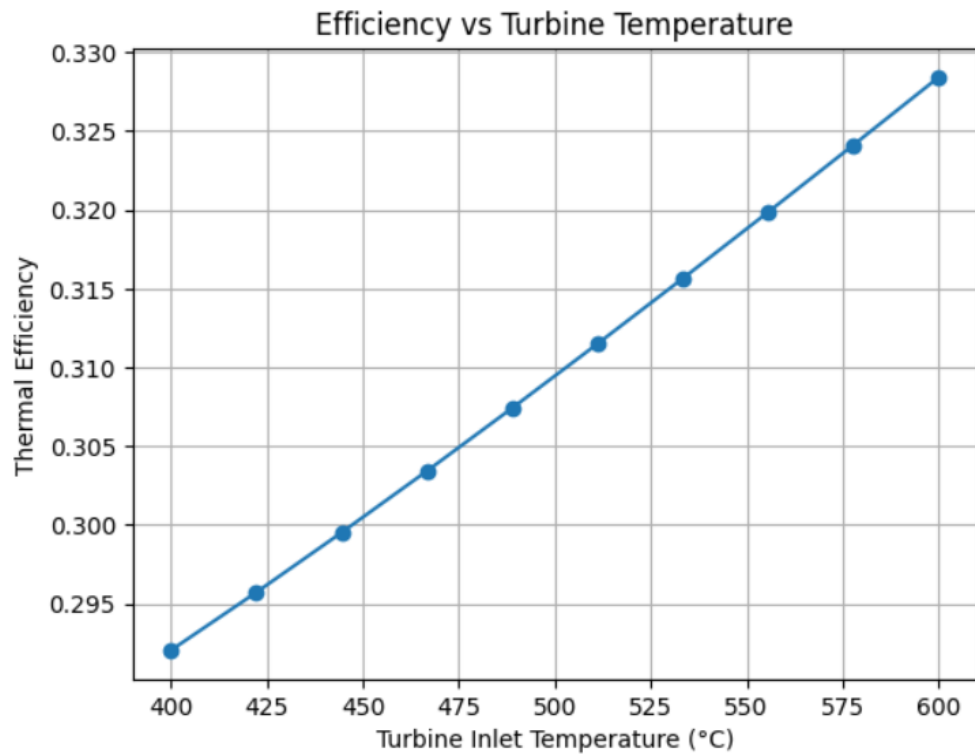


Figure 4: Efficiency vs Temperature

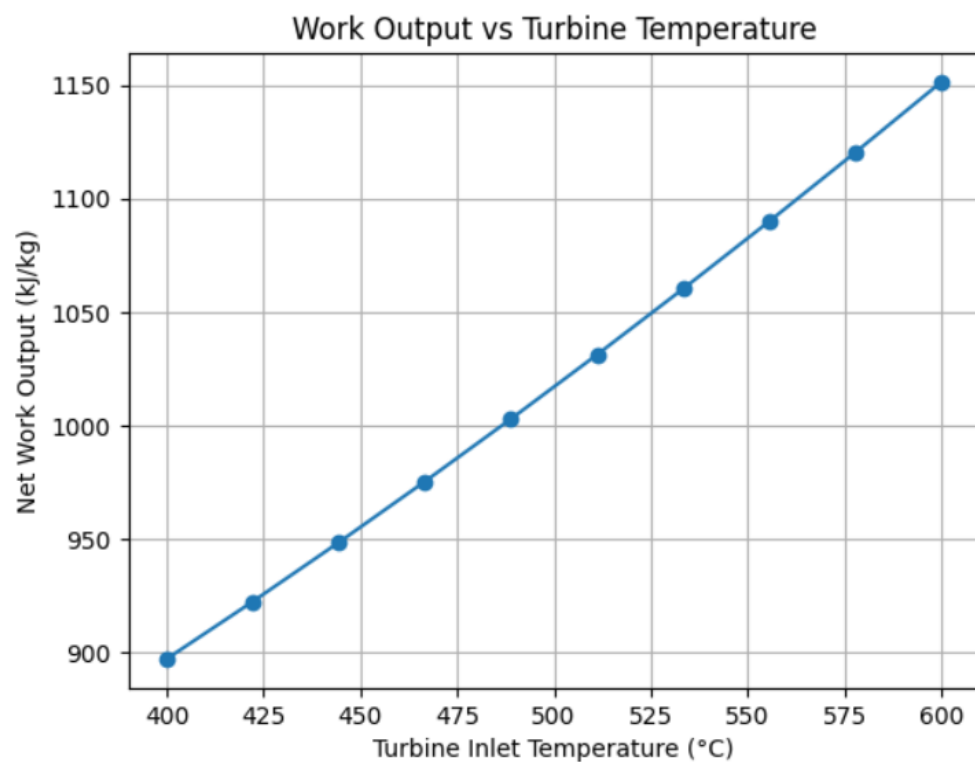


Figure 5: Work vs Temperature



6 Performance Analysis

The analysis shows that the performance of the Rankine cycle improves with increasing operating conditions such as pressure and temperature. Higher boiler pressure enhances the efficiency of the cycle, while higher turbine inlet temperature increases the work output.

The results also help in identifying suitable operating conditions for better performance. However, in practical systems, limitations such as material constraints and safety considerations must also be taken into account.

7 Conclusion

In this project, the performance of the Rankine cycle was studied using a simulation-based approach. The effects of boiler pressure and turbine inlet temperature on thermal efficiency and work output were analyzed.

The results show that increasing boiler pressure and turbine inlet temperature improves the overall performance of the cycle. The simulation provides a clear understanding of how operating conditions influence system behavior.

This study demonstrates the application of thermodynamic principles in analyzing and improving real-world power generation systems.

8 Future Scope

The present study can be extended by incorporating more realistic operating conditions such as component inefficiencies and heat losses. Further analysis can include advanced Rankine cycle configurations such as reheat and regeneration.

Additionally, dynamic modeling of the cycle under varying heat input conditions can be explored to better represent real-world power plant behavior. The integration of renewable energy sources and optimization techniques can also be considered for improving overall system performance.

9 References

- Y. A. Çengel and M. A. Boles, *Thermodynamics: An Engineering Approach*
- Steam property data obtained using the pyXSteam library
- Standard thermodynamics course material on Rankine cycle
- Schematic diagram of Rankine cycle obtained from online sources (Google Images)